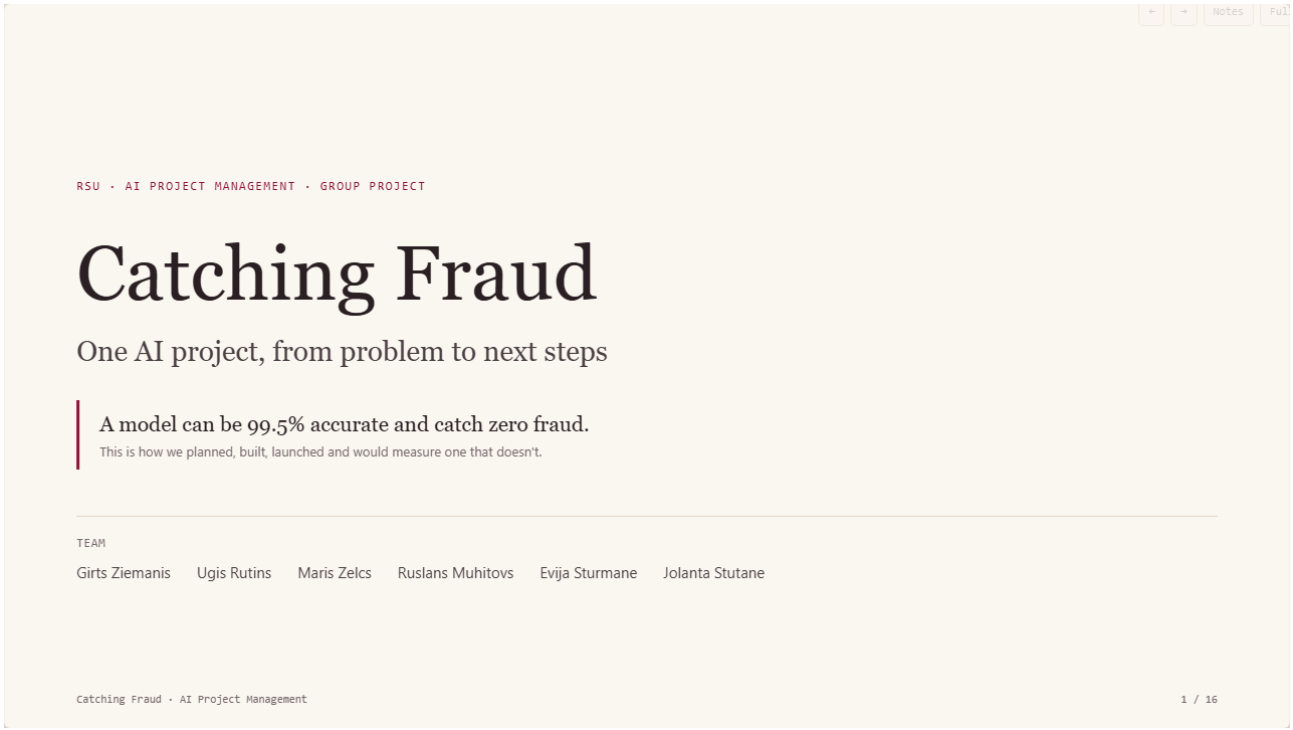
SLIDE 1 / 16 · RSU · AI PROJECT MANAGEMENT · GROUP PROJECT

Title

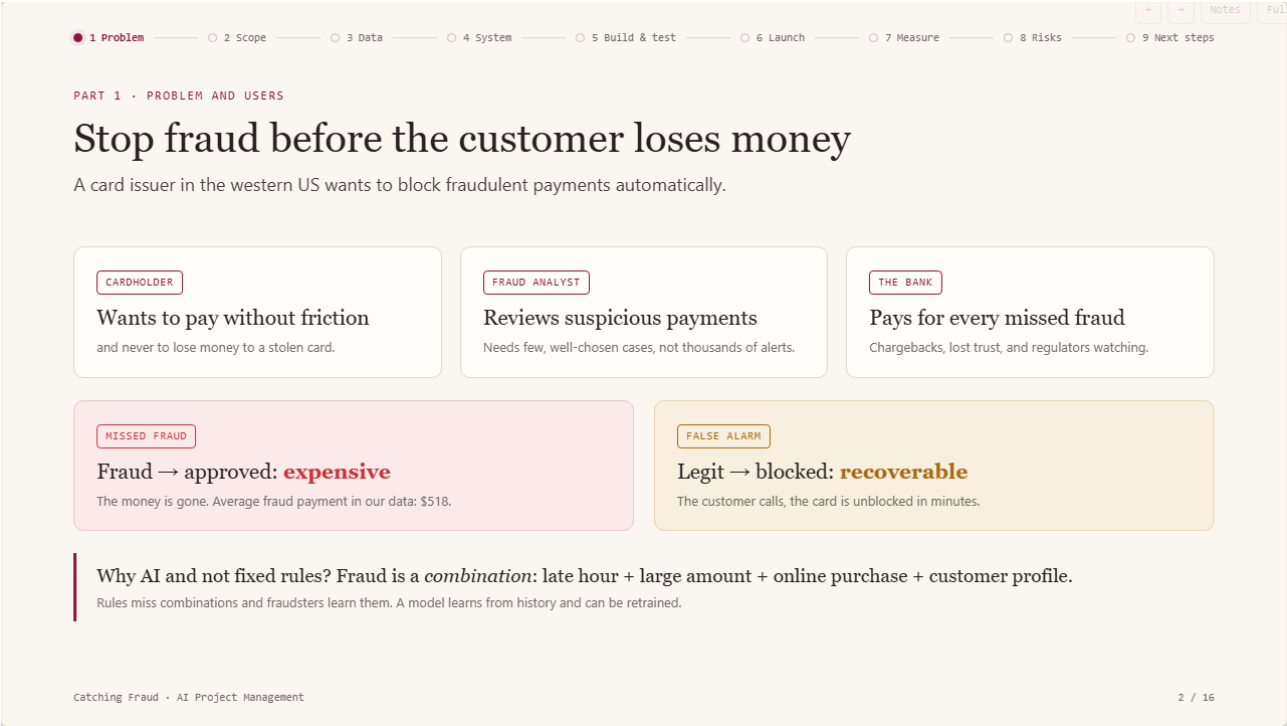


SPEAKER NOTES

- Hook: 99.5% accuracy sounds great, but here it means nothing, because only 0.52% of transactions are fraud.
- We walk through the nine parts of the project brief, from problem and users to next steps. The tracker at the top shows where we are.

SLIDE 2 / 16 · PART 1 · PROBLEM AND USERS

Problem and users

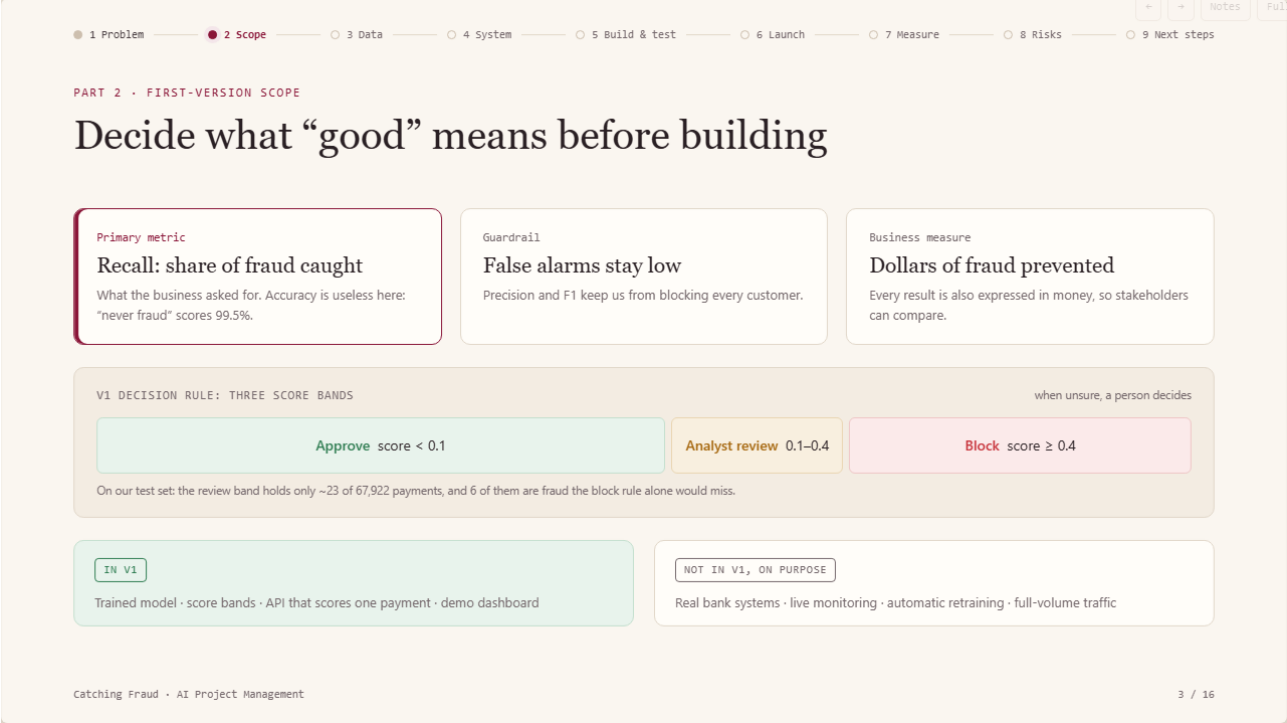


SPEAKER NOTES

- Three users with different needs: the cardholder (no friction, no loss), the fraud analyst (manageable workload), the bank (losses and trust).
- The two mistakes have very different prices. Lecture 6: "a wrong blur is cheap; a missed face is not." Ours: a false alarm is cheap, a missed fraud is not.
- Build or buy: no ready-made public model fits one bank's fraud patterns, so we train our own, with free tools.

SLIDE 3 / 16 · PART 2 · FIRST-VERSION SCOPE

First-version scope

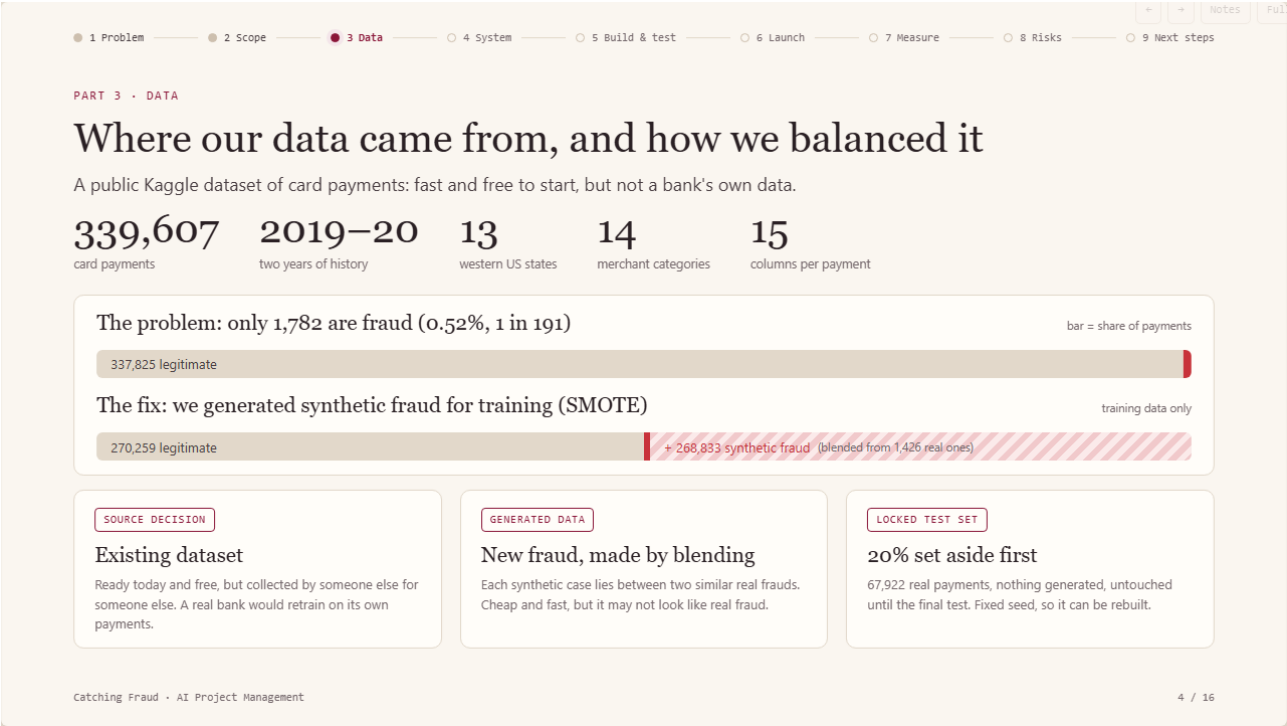


SPEAKER NOTES

- Success criteria were fixed before experimenting: Recall first, false alarms as a guardrail, dollars for stakeholders.
- Lecture 5 "a rule for not sure" and Lecture 6 "when unsure, a person decides": instead of one cutoff we propose three bands.
- Numbers from our threshold analysis: between scores 0.1 and 0.4 there are 6 frauds and 17 legitimate payments in the test set. That is a tiny, high-value workload for analysts. "The AI decides how much work a person gets."
- Scope discipline: the "not in v1" list becomes the roadmap later.

SLIDE 4 / 16 · PART 3 · DATA

Data: where it came from

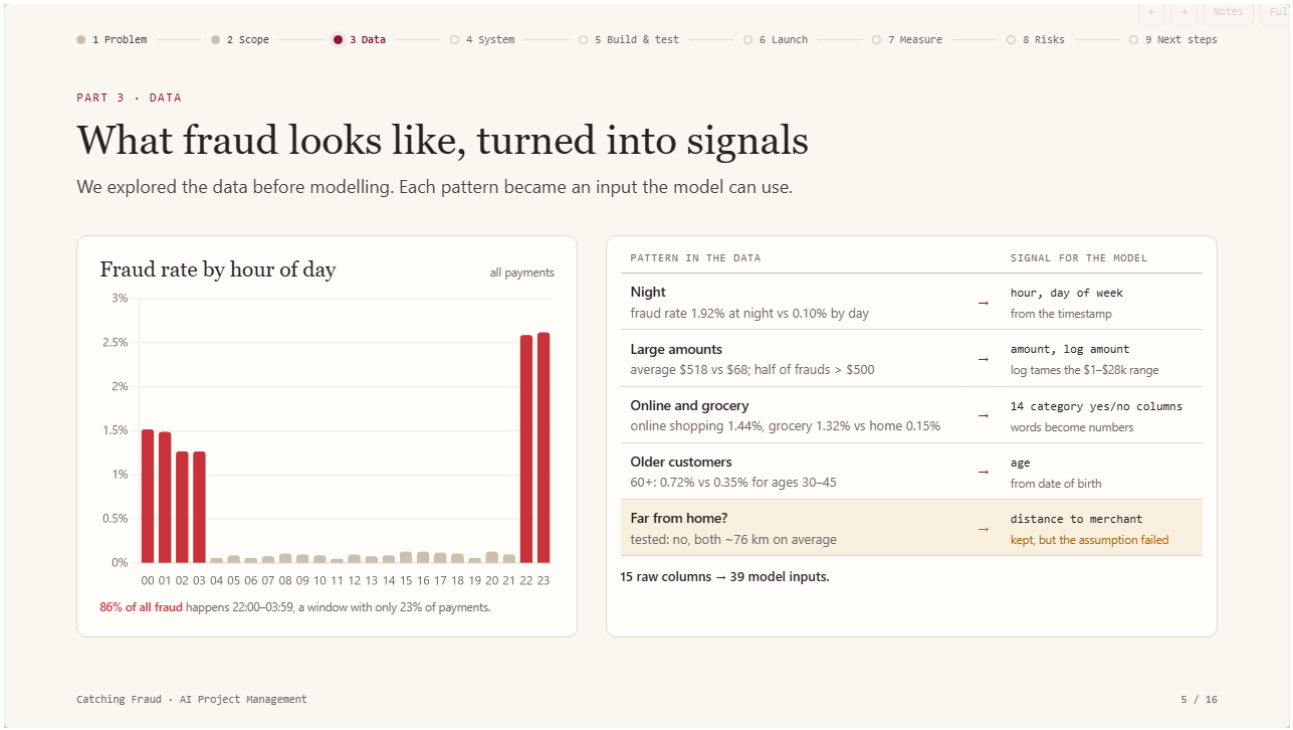


SPEAKER NOTES

- Lecture 2, four data sources: we chose an existing dataset. Cost low, speed "ready today", but fit is often poor: it is not our bank, our customers or our time period.
- The challenge is imbalance: 339,607 payments, but only 1,782 frauds. Lecture 2, "plan the variety, not the number": those few cases decide how good the model can get.
- Fixing the imbalance with generated data (Lecture 2's fourth source: "made by a program; cheap, may not look real"). SMOTE creates a new fraud example on the line between a real fraud and one of its 5 most similar frauds. The training set went from 1,426 frauds among 271,685 payments to a 50/50 balance: 270,259 legitimate and 270,259 fraud, so 268,833 fraud examples are synthetic.
- Only the training data was balanced. The test set stays 100% real, so the final scores reflect real payments.
- We tested two approaches to imbalance: SMOTE for Logistic Regression, Decision Tree and Random Forest; class weights (rare fraud counts more, no generated data) for XGBoost and LightGBM. Part 5 shows what that taught us.
- Labels: here the fraud column is given. In production, the answer comes weeks later through chargebacks (Lecture 8).
- Before any analysis, 20% (67,922 payments) was locked away as the test set, stratified to keep the 0.52% fraud ratio, with random seed 42 so it is reproducible.
- Check whether the Kaggle data is simulated, and say so if it is.

SLIDE 5 / 16 · PART 3 · DATA

What fraud looks like

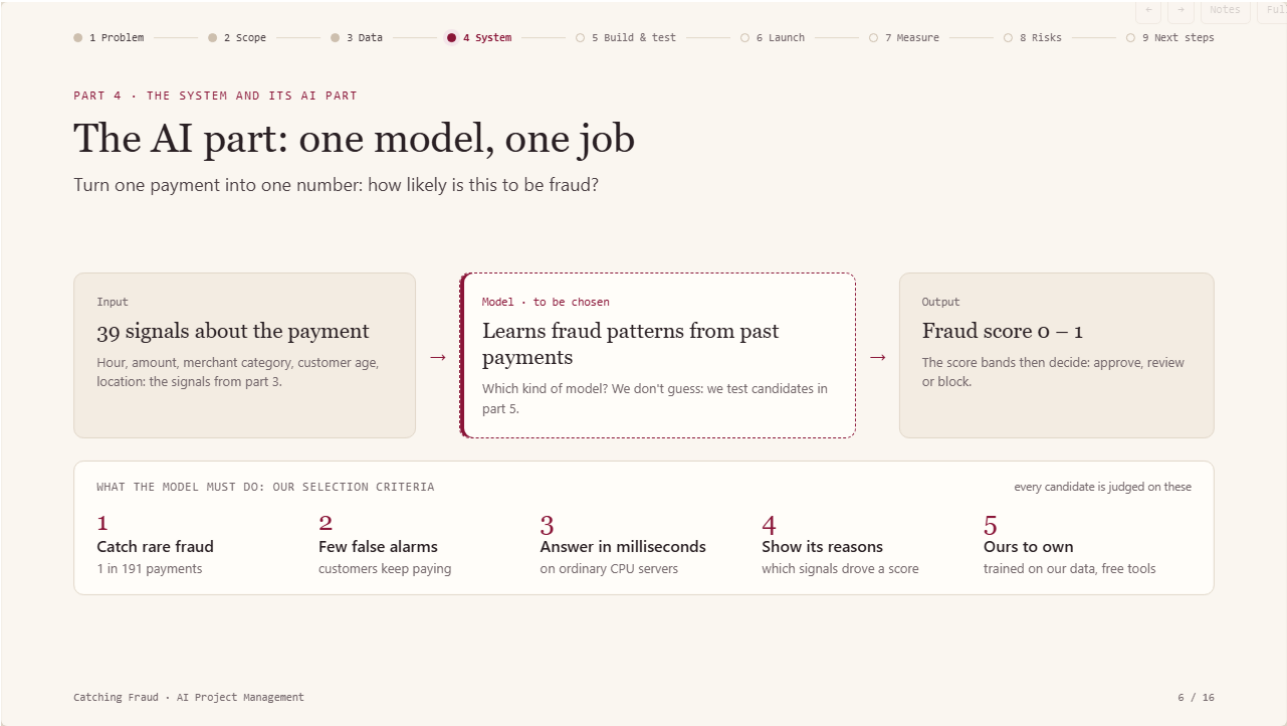


SPEAKER NOTES

- Exploration before modelling: we looked for what makes fraud different, then turned each finding into a signal (feature). All numbers here are computed from the full dataset.
- Night: 86% of fraud happens between 22:00 and 03:59, although only 23% of all payments happen then. Fraud rate at night 1.92% vs 0.10% in the daytime, about 19× higher.
- Amount: average fraud $518 vs $68 for legitimate payments (medians $356 vs $46). 47% of frauds are above $500, and about 1 in 5 payments above $500 is fraud. Log amount compresses the huge range.
- Category: online shopping 1.44%, in-store grocery 1.32%, online misc 1.28%, against 0.15% for home goods. One yes/no column per category (one-hot encoding).
- Age: customers over 60 have a fraud rate of 0.72%, double the 0.35% of 30–45 year olds.
- Tested assumption: we expected fraud far from the cardholder's home. The data says no: average distance is 75.6 km for both. We kept the feature, but this is a project lesson: test assumptions with data instead of trusting intuition.

SLIDE 6 / 16 · PART 4 · THE SYSTEM AND ITS AI PART

The AI part

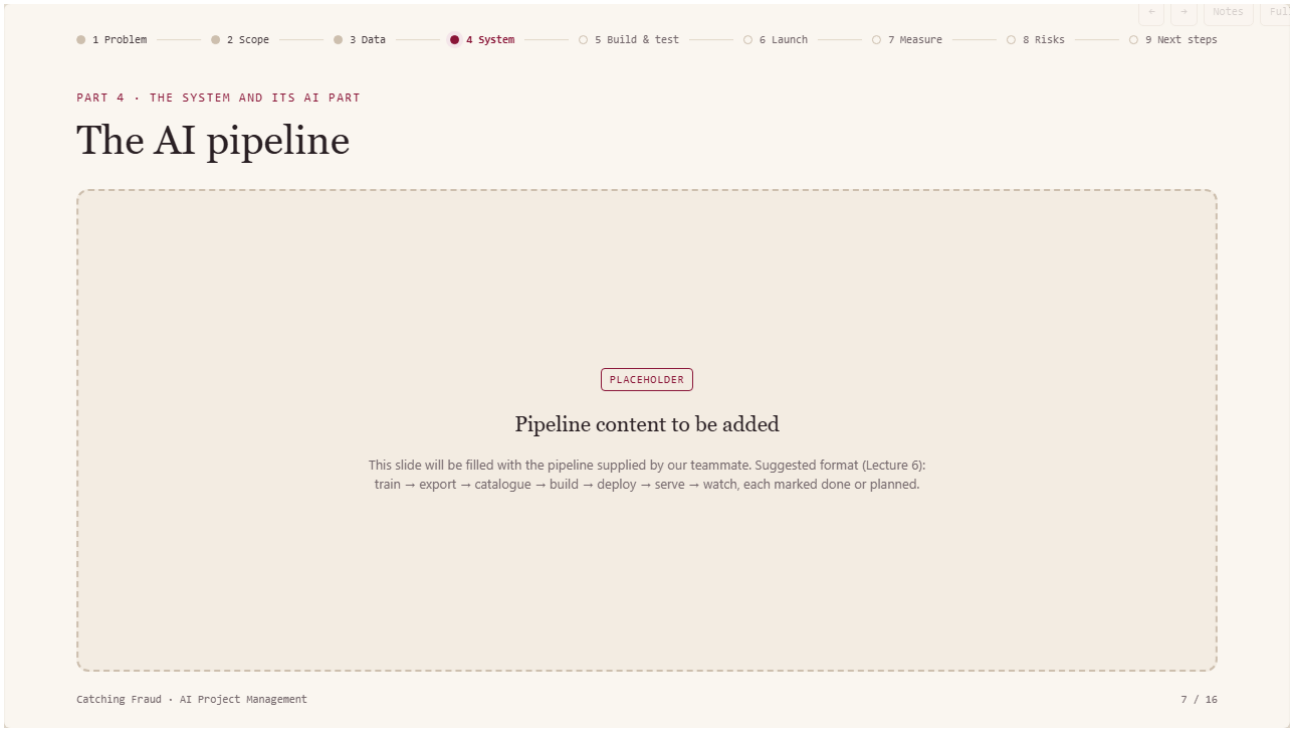


SPEAKER NOTES

- The AI part has one job: give each payment a fraud score from 0 to 1. Everything around it (lookups, features, bands, people) is ordinary programs and people (Lecture 6: one product = AI + programs + people).
- We deliberately don't pick a model here. First we write down what the model must do; then we test candidates against it in part 5.
- Criteria 1 and 2 come from the success criteria in part 2 (Recall, false alarms). Criterion 3 comes from the system: a payment is waiting (Lecture 7). Criterion 4: analysts and customers need reasons. Criterion 5: build or buy (Lecture 6) — no ready-made public model knows one bank's fraud.

SLIDE 7 / 16 · PART 4 · THE SYSTEM AND ITS AI PART

AI pipeline

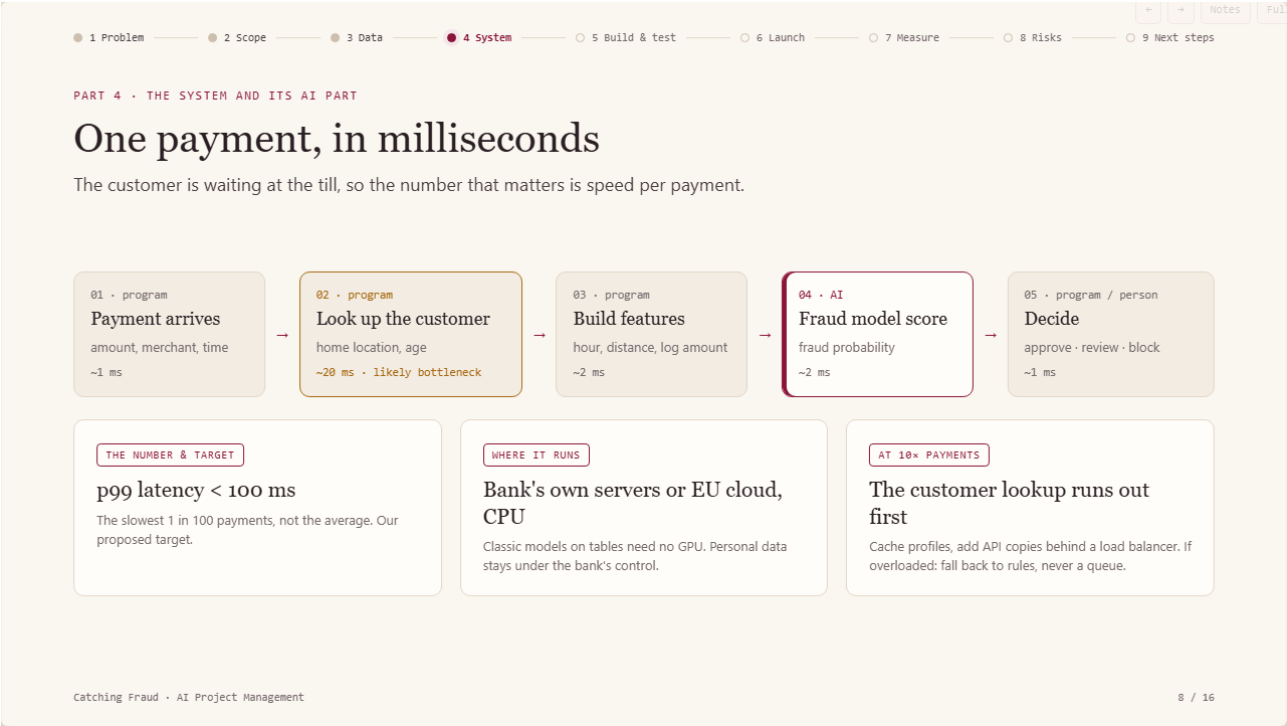


SPEAKER NOTES

- Placeholder: pipeline content and speaker notes to be provided by teammate.

SLIDE 8 / 16 · PART 4 · THE SYSTEM AND ITS AI PART

One payment, in milliseconds

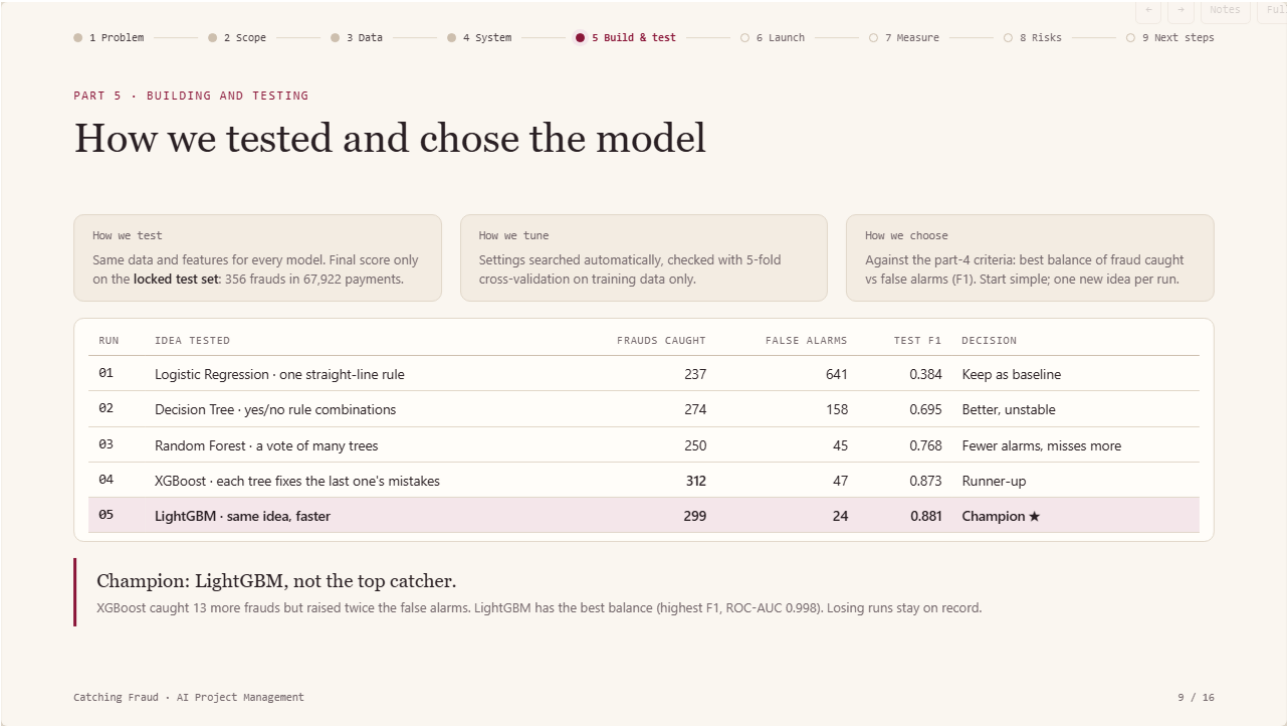


SPEAKER NOTES

- This answers Lecture 7's part-4 questions: draw one request, which number matters with a target, where it runs, what runs out first at 10×.
- Millisecond figures are our planning budget, not measurements. First step in production: measure (Lecture 7: "measure first").
- Lecture 6: "the AI is the clever part, not the heavy part". The model takes a few ms; looking up data around it is the likely bottleneck.
- Lecture 6: one product = AI + programs + people. Only step 4 is AI; step 5 can involve a person (the review band).
- No batching: a payment can't wait for others. Overload plan: "answer smaller" with simple rules.

SLIDE 9 / 16 · PART 5 · BUILDING AND TESTING

Experiment log

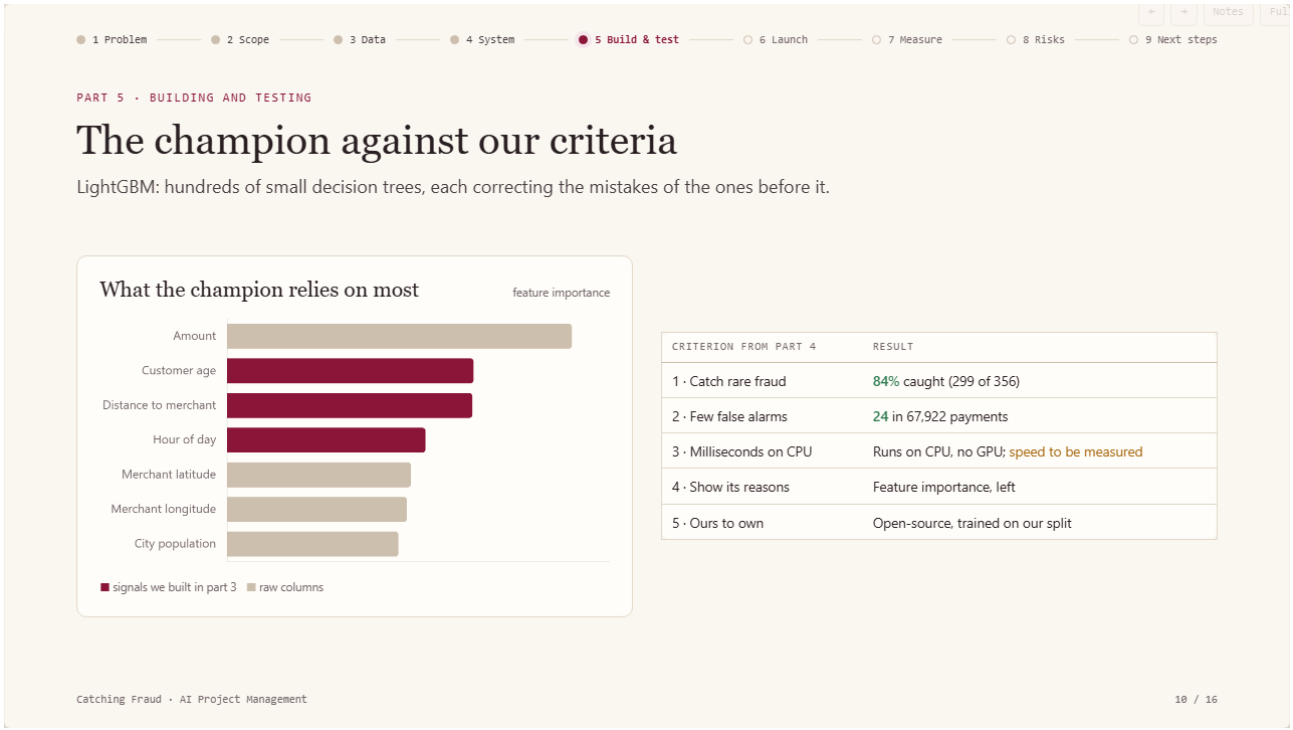


SPEAKER NOTES

- Lecture 3: every run gets a row (idea, change, result, decision), and we change one thing at a time.
- From baseline to champion: 237 → 299 frauds caught (Recall 67% → 84%), false alarms 641 → 24.
- Fair experiment: same split, same features, tuned on F1 with 5-fold cross-validation, judged on the untouched test set.
- The champion choice is a recorded trade-off, not "highest number wins". Lecture 3 champion loop: losers stay on record.

SLIDE 10 / 16 · PART 5 · BUILDING AND TESTING

Inside the champion

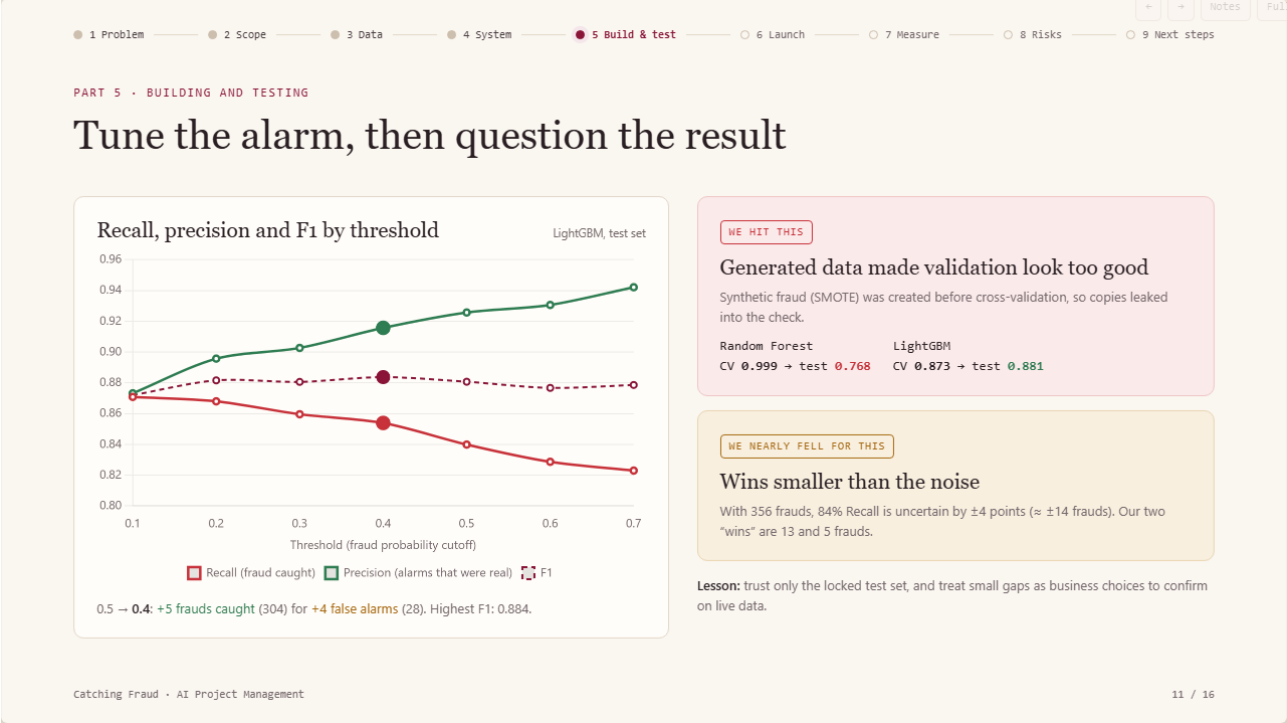


SPEAKER NOTES

- Only now we name the model: after testing five candidates, LightGBM won on the balance of fraud caught and false alarms.
- LightGBM is gradient boosting: many small decision trees built one after another, each fixing the previous ones' mistakes. Trained on the 80% training split (271,685 payments) with class weights for the rare fraud class.
- Feature importance: amount is #1, then age, distance and hour. Three of the top four are signals we built in part 3. Distance ranks high even though fraud and legitimate payments have the same average distance: this importance counts how often a feature is used, not how much it helps, so read it with care.
- Criterion 3 is honest: the model type runs on CPU, but we have not yet measured its speed under load. That's the first step in production (Lecture 7: measure first).

SLIDE 11 / 16 · PART 5 · BUILDING AND TESTING

Tuning and trusting the numbers

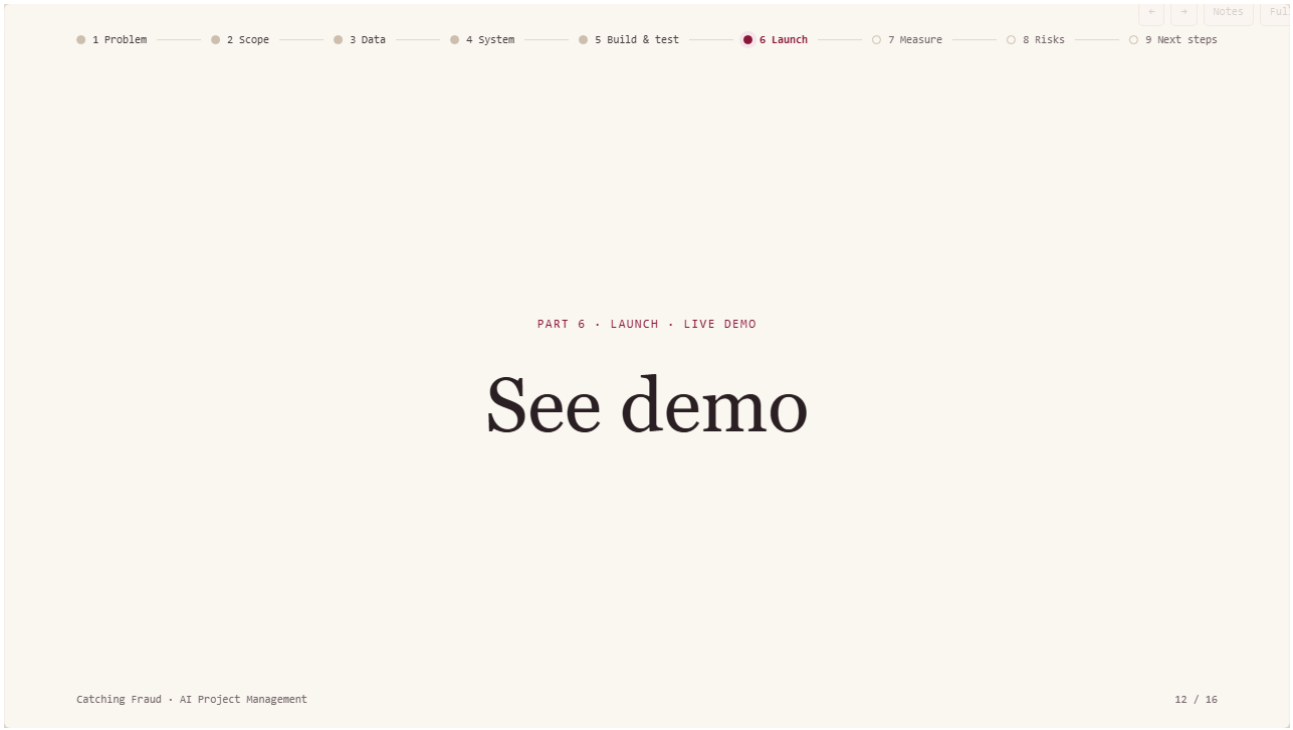


SPEAKER NOTES

- Threshold: the model outputs a probability; the business sets the cutoff. Like a smoke alarm's sensitivity. Optimisation without retraining.
- Lecture 2 warns that generated data "may not look real". SMOTE is generated data: models trained on it scored near-perfect in cross-validation (RF 0.999, DT 0.998, LR 0.979) but much lower on the test set. Class-weighted models (XGBoost, LightGBM) had CV ≈ test. Fix: generate synthetic examples inside each fold.
- Lecture 3 "measure the noise" and Lecture 8 "same AI, three tests, three scores": 95% margin on Recall with 356 cases is about ±3.8 points. LightGBM vs XGBoost (13 frauds) and 0.4 vs 0.5 (5 frauds) sit inside it.
- The final choice still holds, because it was judged on the untouched test set and on the business trade-off.

SLIDE 12 / 16 · PART 6 · LAUNCH · LIVE DEMO

Live demo

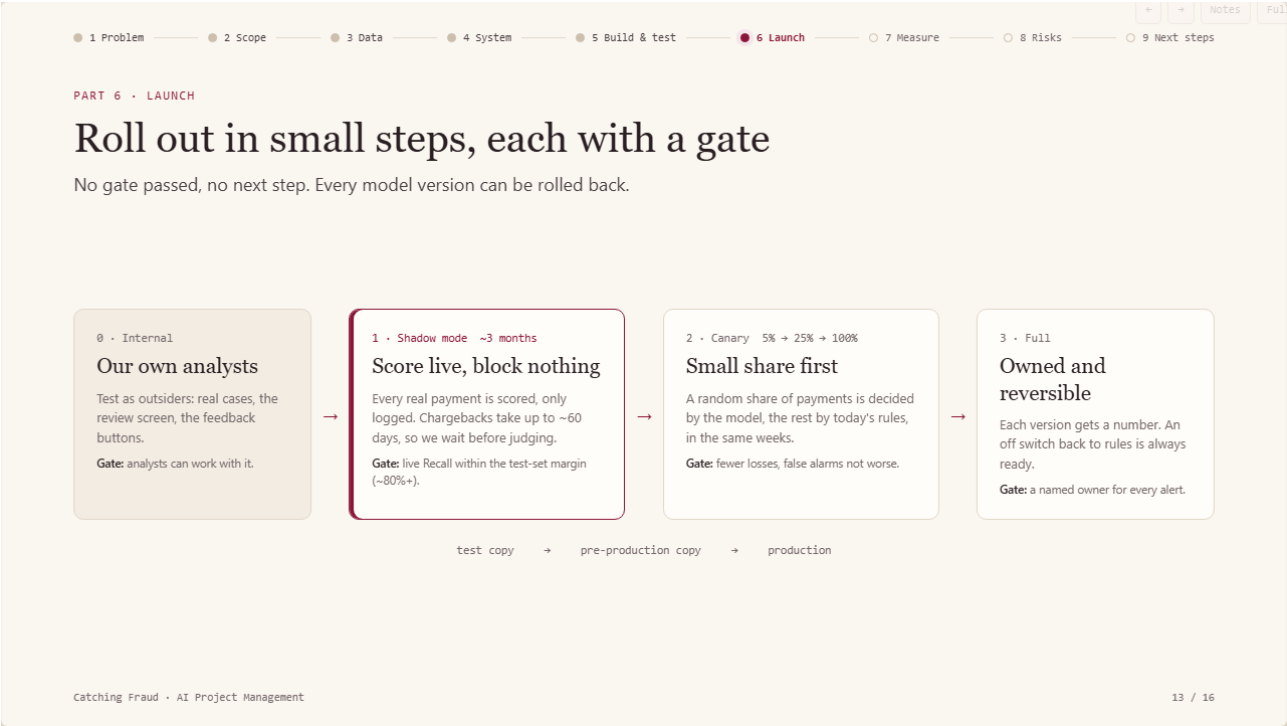


SPEAKER NOTES

- Switch to the demo window (Flask API on port 5050, Next.js dashboard on port 3000).
- Suggested order: an everyday purchase (approved), a payment with the fraud fingerprint (blocked), a borderline case (show how the cutoff decides).
- Backup plan: have screenshots ready in case the demo fails.
- Keep it to about 2 minutes, then come back to the deck.

SLIDE 13 / 16 · PART 6 · LAUNCH

Launch

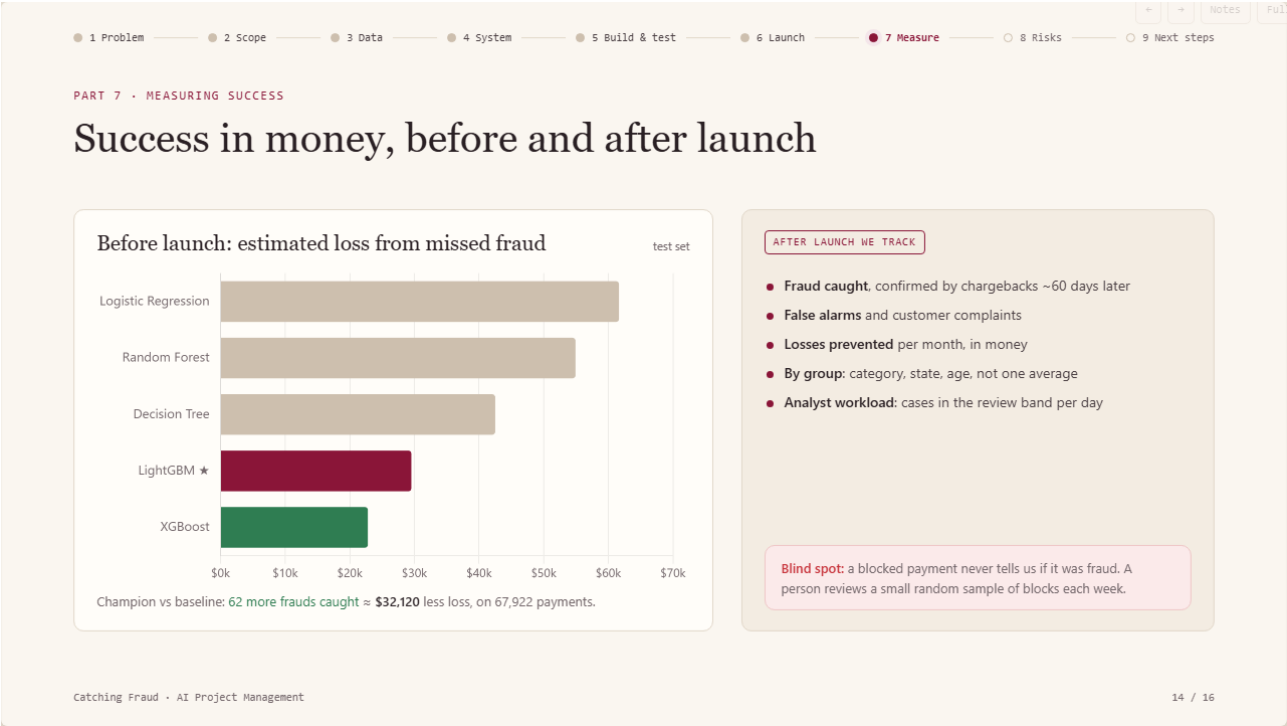


SPEAKER NOTES

- Lecture 5: release to a few people first and test as an outsider. Our first users are our own fraud analysts.
- Lecture 8: the test score is only a promise (SickKids silent trial: 0.90 on old data, 0.50 on new patients). Shadow mode checks the promise on live data without affecting customers. Like Stripe, we must wait for chargebacks before the score is final.
- Canary with an A/B comparison in the same weeks, not before vs after.
- Lecture 6: three copies (test, pre-production, production), version numbers, and going back is normal.
- The ~3 months and ~80% gate are our proposals, to be agreed with the bank.

SLIDE 14 / 16 · PART 7 · MEASURING SUCCESS

Measuring success

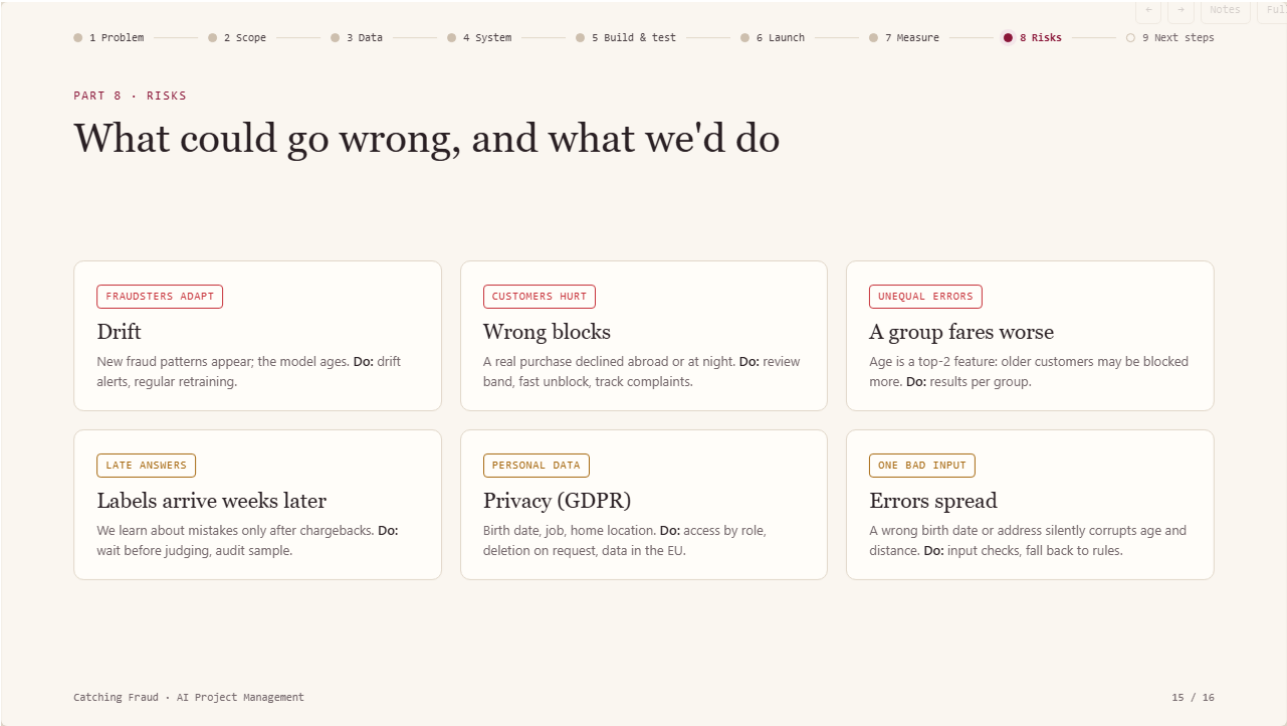


SPEAKER NOTES

- Before launch: Recall translated into money, the business measure from part 2. Estimated losses: Logistic Regression $61,650; Random Forest $54,915; Decision Tree $42,482; LightGBM $29,530; XGBoost $22,795. Assumption: every missed fraud is worth the average $518.
- Lecture 8, "who holds the answer sheet, and when": for us, chargebacks, weeks later.
- Lecture 8, averages that hide a group: report results per category, state and age group.
- Blind spot: blocked payments never get a true label, so false alarms are a guess without a random audit sample.

SLIDE 15 / 16 · PART 8 · RISKS

Risks

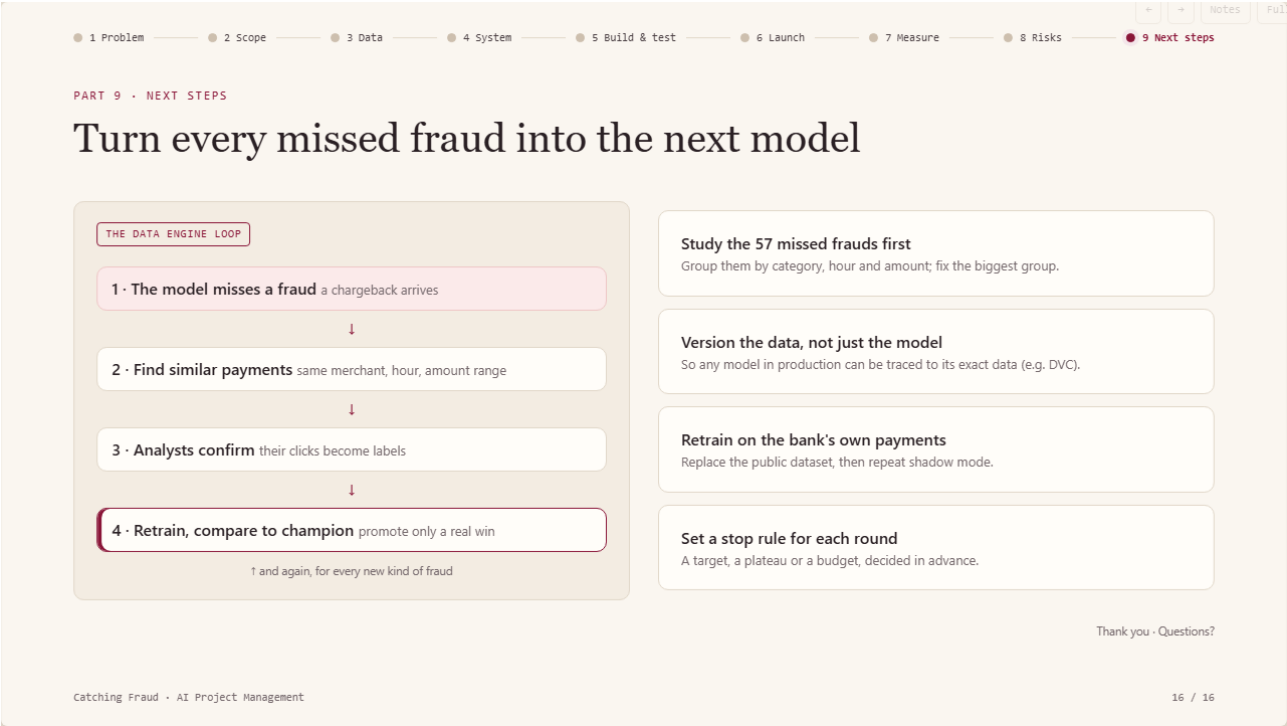


SPEAKER NOTES

- Drift is guaranteed in fraud (Lecture 8: Instacart in 2020). Fraudsters react to what gets blocked.
- Fairness: age is the model's #2 feature. That may be a real fraud signal, but we must check error rates per age group (Lecture 8: averages that hide a group).
- Lecture 2: personal data needs an owner, access by role, and deletion "from every version, following a written process".
- Lecture 6: one wrong answer spreads to many steps. A bad customer profile breaks several features at once.
- From our own project: generated data and small samples can mislead (part 5).

SLIDE 16 / 16 · PART 9 · NEXT STEPS

Next steps



SPEAKER NOTES

- Lecture 2, Tesla's data engine: model fails → find similar cases → label → retrain, repeated for every new failure. Ours starts with a chargeback.
- Lecture 3: the next idea comes from the champion's mistakes, so we start with the 57 frauds LightGBM missed.
- Lecture 6: people's daily work teaches the next model; a person, not a timer, decides when to retrain.
- Lecture 2: "which exact data trained the model?" We save the model, scaler and feature list, but not yet a data version.
- Lecture 3 stop rules: target, plateau or budget.
- Close: the hard part of this AI project was not the algorithm. It was the decisions before and after it.